Two springs in series, one of them damageable: analytical, static and dynamic solutions

Repository `1D_damage_spring_system`

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1 Problem statement

We consider two springs (elastic bars) in series:

$$\text{node 0 (clamped)} \xrightarrow{\text{spring 1, } k_1^d(u_1)} \text{node 1 (mass } m_1\text{)} \xrightarrow{\text{spring 2, } k_2} \text{node 2 (}u_2\text{ imposed)}$$

Spring 1 is damageable, spring 2 remains linear elastic. The loading is displacement controlled: u_2 is imposed at node 2, increasing from 0 to u_2^f. The unknown is the displacement u_1 of the intermediate node. Spring 1 softens with damage, which produces a *snap-back*: the equilibrium branch $F(u_2)$ turns back in u_2, and quasi-static displacement control cannot follow it. We compare:

1. the **analytical** solution (closed form, complete branch including the unstable part);
2. the **static** solution (Newton–Raphson at each u_2 increment);
3. the **dynamic** solution (mass m_1 at node 1, implicit Newmark scheme, Newton–Raphson at each time step).

The focus of the comparison is the shape of the residual $R(u_1)$ that the Newton–Raphson has to cancel, and the way the inertia term regularises its non-monotonicity.

1.1 Damage law

The damage of spring 1 is a variable $d \in [0, 1]$. Its trial value is a function of the current elongation (tension only), and irreversibility is enforced on the damage itself: the damage at time t is the largest trial value reached so far,

$$\hat{d}(u_1) = 1 - \exp\left(-\frac{\langle u_1\rangle_+^2}{u_c^2}\right), \qquad d(t) = \max_{\tau \le t} \hat{d}(u_1(\tau)), \qquad k_1^d = k_1\,(1-d), \qquad F_1 = k_1^d\,u_1. \quad (1)$$

On loading ($\hat{d}(u_1) \ge d$), $d = \hat{d}(u_1)$ and $d'(u_1) = \dfrac{2u_1}{u_c^2}\exp(-u_1^2/u_c^2)$; on unloading ($\hat{d}(u_1) < d$), d is frozen and the spring unloads elastically with the secant stiffness $k_1(1-d)$. Since $\hat{d}$ is monotonic in $u_1 \ge 0$, this is equivalent to a history variable $\kappa = \max u_1$ with $d = \hat{d}(\kappa)$; the formulation on d is the one used in the code (`model.damage_history`). Figure 1 shows $d(u_1)$, k_1^d/k_1 and $F_1(u_1)$.

1.2 Parameters

Consistent unit system N / mm / s, hence mass in tonne (1 t = 1000 kg).

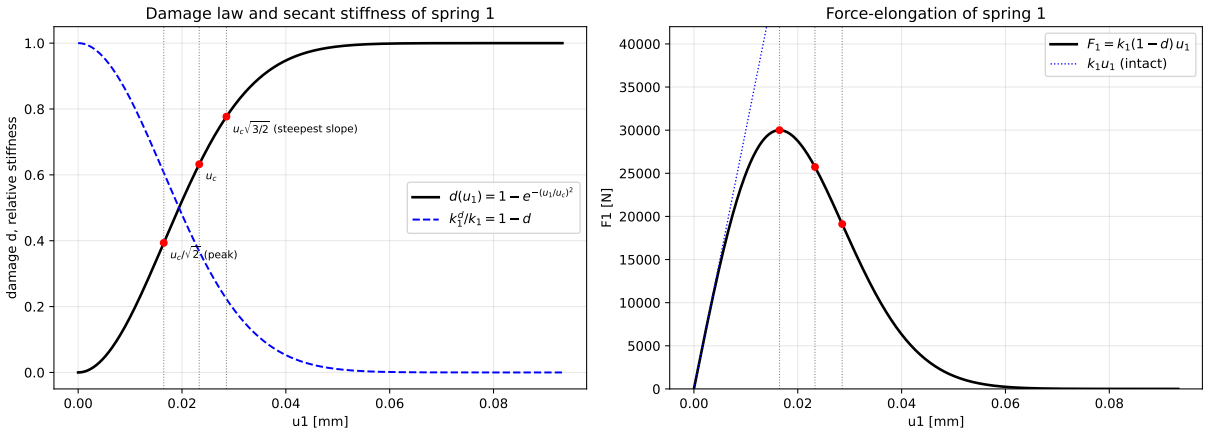


Figure 1: Damage law of spring 1. Left: $d(u_1)$ and the relative secant stiffness $1-d$; right: force $F_1 = k_1(1-d)u_1$. The marks are the force peak $u_1 = u_c/\sqrt{2}$ ($d = 1-e^{-1/2}$), $u_1 = u_c$ ($d = 1-e^{-1}$) and the point of steepest descent $u_1 = u_c\sqrt{3/2}$ where $\mathrm{d}F_1/\mathrm{d}u_1 = -2e^{-3/2}k_1$.

Spring 1	$E_1 = 10\,000$ MPa, $A_1 = 20$ mm^2, $L_1 = 5$ mm $u_c = 5$ mm	$k_1 = E_1A_1/L_1 = 40\,000$ N/mm
Spring 2	$E_2 = 20\,000$ MPa, $A_2 = 30$ mm^2, $L_2 = 500$ mm	$k_2 = E_2A_2/L_2 = 1\,200$ N/mm
Mass	$m_1 = 10$ t (node 1)	
Loading	$u_2 : 0 \to u_2^f = 80$ mm	
Newton–Raphson	tolerance $\lvert R\rvert \le 10^{-6}$ N, 50 iterations max	

The ratio $k_1/k_2 = 33.3$: spring 1 is very stiff while intact, and almost all of the imposed displacement is absorbed by spring 2 ($u_1 \simeq u_2\,k_2/(k_1+k_2)$). The mass $m_1 = 10$ t is not physical for a $20\,\mathrm{mm}^2 \times 5$ mm bar (actual mass $\rho A_1 L_1 \sim 10^{-6}$ t): it is a numerical regularisation parameter, see §5.

2 From the mass–spring system to the residuals

2.1 Equation of motion of node 1

The forces acting on the mass m_1 are the force of spring 1, $f_1 = k_1(1-d)\,u_1$, directed towards node 0, and the force of spring 2, $f_2 = k_2(u_2 - u_1)$, directed towards node 2. Newton's second law gives

$$m_1\ddot{u}_1 = -k_1(1-d)\,u_1 + k_2(u_2(t) - u_1) \quad\Longleftrightarrow\quad m_1\ddot{u}_1 + \underbrace{(k_1^d(u_1) + k_2)u_1}_{f^{\mathrm{int}}(u_1)} - \underbrace{k_2\,u_2(t)}_{f^{\mathrm{ext}}(t)} = 0. \qquad (2)$$

In the notation of the companion note on nonlinear dynamics, $M = m_1$, $C = 0$, $f^{\mathrm{int}}(u_1) = (k_1^d + k_2)u_1$ and $f^{\mathrm{ext}}(t) = k_2u_2(t)$: the imposed displacement enters as an external force. This is the usual finite element treatment of a prescribed degree of freedom. Writing the two-node system

$$\begin{bmatrix} k_1^d + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ F_2 \end{Bmatrix}, \qquad (3)$$

the first row is the equilibrium of node 1 with the known term k_2u_2 moved to the right-hand side, and the second row gives the reaction $F_2 = k_2(u_2 - u_1)$, computed after the solve.

2.2 Damage law and its derivative

With $x = u_1/u_c$ and $d = \hat{d}(u_1)$ on loading,

$$k_1^d(u_1) = k_1 e^{-x^2}, \qquad \frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} = -\frac{2k_1 u_1}{u_c^2} e^{-x^2}, \qquad \frac{\mathrm{d}F_1}{\mathrm{d}u_1} = k_1^d + u_1 \frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} = k_1 e^{-x^2}(1 - 2x^2). \tag{4}$$

On unloading ($\hat{d}(u_1) < d$), k_1^d is constant and its derivative is zero.

2.3 Dynamic residual (Newmark)

Substituting the Newmark relation $\ddot{u}_1^{n+1} = c_1(u_1^{n+1} - \tilde{u})$, $c_1 = 1/(\beta \Delta t^2)$, into (2) at time t^{n+1} gives the residual and tangent of the dynamic problem:

$$r_{\mathrm{dyn}}(u_1) = m_1 c_1 \,(u_1 - \tilde{u}) + (k_1^d(u_1) + k_2)u_1 - k_2 u_2^{n+1}, \qquad \frac{\partial r_{\mathrm{dyn}}}{\partial u_1} = m_1 c_1 + k_t(u_1), \tag{5}$$

with k_t the static tangent defined below.

2.4 Static residual and Newton–Raphson

The quasi-static case is the limit $m_1 \ddot{u}_1 \to 0$ (equivalently $m_1 = 0$ in (5)):

$$R(u_1; u_2) = (k_1^d(u_1) + k_2)u_1 - k_2 u_2 = 0, \tag{6}$$

$$k_t(u_1) = \frac{\partial R}{\partial u_1} = k_1^d + u_1 \frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} + k_2 = k_1 e^{-x^2}(1 - 2x^2) + k_2 \quad \text{(loading)}. \tag{7}$$

Newton–Raphson: $u_1^{(i+1)} = u_1^{(i)} - R(u_1^{(i)})/k_t(u_1^{(i)})$, starting from the solution of the previous increment. The dynamic residual (5) is simply R augmented by $m_1 c_1 (u_1 - \tilde{u})$.

2.5 Snap-back condition and multiplicity of roots

R is non-monotonic in u_1 as soon as k_t changes sign. The steepest negative slope of spring 1 is obtained from $\mathrm{d}^2 F_1/\mathrm{d}u_1^2 = 0$, i.e. $x^2 = 3/2$, where $\mathrm{d}F_1/\mathrm{d}u_1 = -2e^{-3/2}k_1$. Hence

$$\text{snap-back} \iff k_2 < 2e^{-3/2}k_1 \simeq 0.446\, k_1. \tag{8}$$

In that case $k_t < 0$ on an interval $[u_1^{\lim}, u_1^{\lim\prime}]$, the residual has up to three roots for a given u_2, and the Newton–Raphson started just before the local maximum of R cycles. Inertia adds $m_1 c_1$ to the tangent: if $m_1 c_1 > -\min k_t$, r_{dyn} is strictly increasing and has a single root.

2.6 Closed-form branch

R is linear in u_2, so the equilibrium branch is obtained by parametrising with u_1 (monotonic loading, $d = \hat{d}(u_1)$):

$$u_2(u_1) = u_1\Big(1 + \frac{k_1}{k_2} e^{-x^2}\Big), \qquad F(u_1) = k_2(u_2 - u_1) = k_1 u_1 e^{-x^2}. \tag{9}$$

- **Force peak**: $\mathrm{d}F/\mathrm{d}u_1 = 0 \Rightarrow u_1^{\mathrm{peak}} = u_c/\sqrt{2}$, $F_{\max} = k_1 u_c/\sqrt{2e}$, $d^{\mathrm{peak}} = 1 - e^{-1/2} = 0.393$. This is what sets u_c from the tensile strength: $u_c = \sqrt{2e}\, f_t A_1/k_1$.
- **Limit point** (turning point of u_2): $\mathrm{d}u_2/\mathrm{d}u_1 = k_t/k_2 = 0$, root of $e^{-x^2}(2x^2 - 1) = k_2/k_1$ on $x \in [1/\sqrt{2}, \sqrt{3/2}]$. Displacement control cannot go beyond $u_2^{\lim}$; it jumps at fixed u_2 to the root on the damaged branch.
- **Damaged branch**: for $x \gg 1$, $k_1^d \to 0$, $u_2 \to u_1$, $F \to 0$.

- **Energies**: free energy $\Psi = \frac{1}{2}k_1(1-d)u_1^2 + \frac{1}{2}k_2(u_2-u_1)^2$, energy release rate $Y = \frac{1}{2}k_1u_1^2$, dissipation $\mathcal{D} = \int Y\,\mathrm{d}d$. Along the branch $\int F\,\mathrm{d}u_2 = \Psi_2 - \Psi_1 + \mathcal{D}_{\text{branch}}$; at the jump ($u_2$ constant, zero work) the released energy $\Psi_1 - \Psi_2 = \mathcal{D}_{\text{branch}} + |W|$ becomes kinetic energy. The total fracture energy is $\int_0^\infty F\,\mathrm{d}u_1 = \frac{1}{2}k_1u_c^2$, i.e. $G_f = k_1u_c^2/2A_1$.
- **Free vibration after rupture**: $m_1\ddot{u}_1 + k_2(u_1 - u_2^f) = 0$, period $2\pi\sqrt{m_1/k_2}$, amplitude $u_2^{\lim} - u_1^{\lim}$ without damping.

3 Analytical solution

The equilibrium of node 1 reads

$$R(u_1;u_2) = k_1^d(u_1)\,u_1 - k_2\,(u_2-u_1) = u_1(k_1^d(u_1)+k_2) - k_2u_2 = 0. \tag{10}$$

Rather than solving (10) for u_1 at given u_2 (a transcendental equation with several roots), we parametrise the equilibrium branch by u_1:

$$u_2(u_1) = u_1\left(1 + \frac{k_1^d(u_1)}{k_2}\right), \qquad F(u_1) = k_2\,(u_2-u_1) = k_1^d(u_1)\,u_1 = k_1\,u_1\,e^{-u_1^2/u_c^2}. \tag{11}$$

With $x = u_1/u_c$, the tangent stiffness of spring 1 is $\mathrm{d}F/\mathrm{d}u_1 = k_1e^{-x^2}(1-2x^2)$ and the system tangent is $k_t(u_1) = k_1e^{-x^2}(1-2x^2) + k_2$. Hence:

- **force peak** at $x = 1/\sqrt{2}$: $u_1 = u_c/\sqrt{2} = 3.536$ mm, $F_{\max} = k_1u_ce^{-1/2}/\sqrt{2} = 85\,776$ N, $u_2 = 75.02$ mm;
- **limit point** (snap-back) where $\mathrm{d}u_2/\mathrm{d}u_1 = k_t/k_2 = 0$: $u_1 = 3.624$ mm, $u_2^{\lim} = 75.06$ mm; beyond it, u_2 decreases along the equilibrium branch down to $u_2 \simeq 13.3$ mm and then rises again on the fully damaged branch ($d \to 1$, $F \to 0$, $u_2 \simeq u_1$).

Quasi-static displacement control therefore cannot go beyond $u_2^{\lim}$ without jumping, at fixed u_2, from the state ($u_1 = 3.62,\ F = F_{\max}$) to the state ($u_1 \simeq u_2,\ F \simeq 0$).

Energy balance of the jump. With the free energy $\Psi = \frac{1}{2}k_1(1-d)u_1^2 + \frac{1}{2}k_2(u_2-u_1)^2$ and the damage dissipation $\mathcal{D} = \int Y\,\mathrm{d}d$, $Y = \frac{1}{2}k_1u_1^2$, the work of the force along the equilibrium branch between the two states of equal u_2 satisfies $W = \int F\,\mathrm{d}u_2 = \Psi_2 - \Psi_1 + \mathcal{D}_{\text{branch}}$. Since the jump occurs at constant u_2 ($W = 0$), the released energy is $\Psi_1 - \Psi_2 = \mathcal{D}_{\text{branch}} + |W|$. Numerically: $\Psi_1 - \Psi_2 = 3\,217$ J, $\mathcal{D}_{\text{branch}} = 455$ J, $|W| = 2\,762$ J. The area $|W|$ between the vertical jump and the equilibrium branch is thus the energy released as kinetic energy at the jump, in excess of what the quasi-static branch would have dissipated by damage. (The script calls it `released_energy`.)

4 Static solution

u_2 is imposed over 2000 uniform increments and (10) is solved by Newton–Raphson with the analytical tangent

$$k_t(u_1) = k_1^d(u_1) + u_1\,\frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} + k_2, \qquad u_1^{(i+1)} = u_1^{(i)} - \frac{R(u_1^{(i)})}{k_t(u_1^{(i)})}, \tag{12}$$

starting from the solution of the previous increment. The residual (10) is not monotonic in u_1 ($k_t < 0$ for $u_1 \in [3.62\,;\simeq 12]$ mm, minimum $k_t = -16\,650$ N/mm): at a given u_2 it may have one or three roots (figure 3, left).

Behaviour at the limit point. For $u_2 > u_2^{\lim}$ there is no longer a root in the neighbourhood of the previous iterate: the Newton–Raphson behaves erratically. With fine increments it cycles around the local maximum of the residual, whose value $R(u_1^{\lim}; u_2) = -k_2(u_2 - u_2^{\lim})$ is negative, and never converges. With coarse increments (the `inc` used here), an iterate may land very close to that maximum, where the tangent k_t is almost zero: the Newton step $-R/k_t$ is then huge and may throw the iterate beyond the softening zone, onto the fully damaged branch where the residual is the straight line $k_2(u_1 - u_2)$; one more iteration lands on $u_1 = u_2$ and the Newton–Raphson "jumps" by itself. The static curve then drops vertically from $F_{\max}$ to zero at that increment (figure 4). Which of the two behaviours occurs depends on the increment size in an unpredictable way, and the jump, when it happens, occurs at the first increment beyond $u_2^{\lim}$ rather than at $u_2^{\lim}$ itself. No numerical device is used (no restart on the damaged branch, no arc-length): if the Newton–Raphson fails, the quasi-static calculation stops at that increment with a message. The dynamic solution of §5 is the physical continuation. Away from the limit point, a few iterations suffice.

5 Dynamic solution

A mass m_1 is added at node 1; node 2 remains displacement controlled by a ramp $u_2(t) = v_2\, t$ up to u_2^f, then held at u_2^f for n_{hold} steps to let the jump complete. The equation of motion of node 1 (undamped) is

$$m_1 \ddot{u}_1 + k_1^d(u_1)\, u_1 - k_2\,(u_2(t) - u_1) = 0. \tag{13}$$

Newmark scheme. Implicit Newmark, $\beta = \frac{1}{4}$, $\gamma = \frac{1}{2}$ (average acceleration, unconditionally stable, no numerical damping), in predictor–corrector form:

$$\tilde{u} = u_n + \Delta t\, v_n + \tfrac{\Delta t^2}{2}(1 - 2\beta)\, a_n, \qquad \tilde{v} = v_n + (1 - \gamma)\Delta t\, a_n, \tag{14}$$

$$a_{n+1} = c_1\,(u_{n+1} - \tilde{u}), \quad c_1 = \frac{1}{\beta \Delta t^2}, \qquad v_{n+1} = \tilde{v} + \gamma \Delta t\, a_{n+1}. \tag{15}$$

For a general damped system $M\ddot{u} + C\dot{u} + f^{\text{int}}(u) - f^{\text{ext}}(t) = 0$, substituting both Newmark relations gives the residual and its tangent

$$r(u^{n+1}) = \Big(\frac{M}{\beta \Delta t^2} + \frac{C\gamma}{\beta \Delta t}\Big)(u^{n+1} - \tilde{u}) + C\tilde{v} + f^{\text{int}}(u^{n+1}) - f^{\text{ext}}(t^{n+1}), \tag{16}$$

$$\frac{\partial r}{\partial u} = \frac{M}{\beta \Delta t^2} + \frac{C\gamma}{\beta \Delta t} + \frac{\partial f^{\text{int}}}{\partial u}. \tag{17}$$

Here $C = 0$, so the residual to cancel at step $n+1$ and its tangent are those of the static case augmented by the inertia term:

$$R_{\text{dyn}}(u_1) = m_1 c_1\,(u_1 - \tilde{u}) + R(u_1; u_2^{n+1}), \qquad k_t^{\text{dyn}}(u_1) = m_1 c_1 + k_t(u_1). \tag{18}$$

The Newton–Raphson is initialised with the predictor $\tilde{u}$. After convergence, a_{n+1} and v_{n+1} are updated with the corrector relations; at $t = 0$, $a_0 = (f^{\text{ext}}(0) - f^{\text{int}}(u_0))/m_1$. The history variable d is frozen during the iterations of a step and updated after convergence.

Choice of the time step: inertial regularisation. From (18), inertia adds to the tangent the constant $m_1 c_1 = m_1/(\beta \Delta t^2)$. If $m_1 c_1 > -\min k_t = 16\,650$ N/mm, the dynamic residual is strictly increasing in u_1: a single root, unambiguous Newton–Raphson, no more abrupt jump at the limit point. With the natural pulsation of the intact system $\omega_0 = \sqrt{(k_1 + k_2)/m_1} = 64.2$ rad/s ($T_0 = 97.9$ ms) and $\Delta t = T_0/N$, the inertia/stiffness ratio is

$$\frac{m_1 c_1}{k_1 + k_2} = \frac{N^2}{4\pi^2 \beta} = \frac{N^2}{\pi^2} \quad (\beta = \tfrac{1}{4}). \tag{19}$$

A fine step ($N = 40$, initial choice) gives a ratio $\simeq 160$: the dynamic residual is an almost perfect straight line and the comparison with statics loses its interest. We keep $N = 5$, i.e. $\Delta t = 19.6$ ms, $m_1 c_1 = 104\,360$ N/mm, ratio 2.5: enough to regularise (minimum dynamic tangent +87 700 N/mm) without flattening the contribution of the damaged spring. The ramp lasts 150 steps (2.94 s, i.e. 30 natural periods: loading slow compared with the system dynamics), followed by 100 hold steps. The Newton–Raphson converges in at most 3 iterations.

Results. The dynamic solution follows the analytical equilibrium branch beyond the limit point (u_2 up to $\simeq 78$ mm before F drops, figure 2), then node 1 "jumps" to the damaged branch during the hold. Since the released energy (§3) has no dissipation mechanism (no damping, spring 1 already fully damaged), the mass then oscillates freely on spring 2 about $u_1 = u_2^f$ with period $2\pi\sqrt{m_1/k_2} = 0.574$ s and amplitude $\simeq 73$ mm (figure 5); F_2 then becomes alternately negative (spring 2 in compression). Thanks to irreversibility ($d = \max \hat{d}$), damage stays at $d = 1$ during these oscillations; with the reversible law $d = d(u_1)$ of the original scripts, d dropped back to 0.66 at the low point ($u_1 \simeq 5$ mm) and spring 1 "healed", which is non-physical.

6 Figures

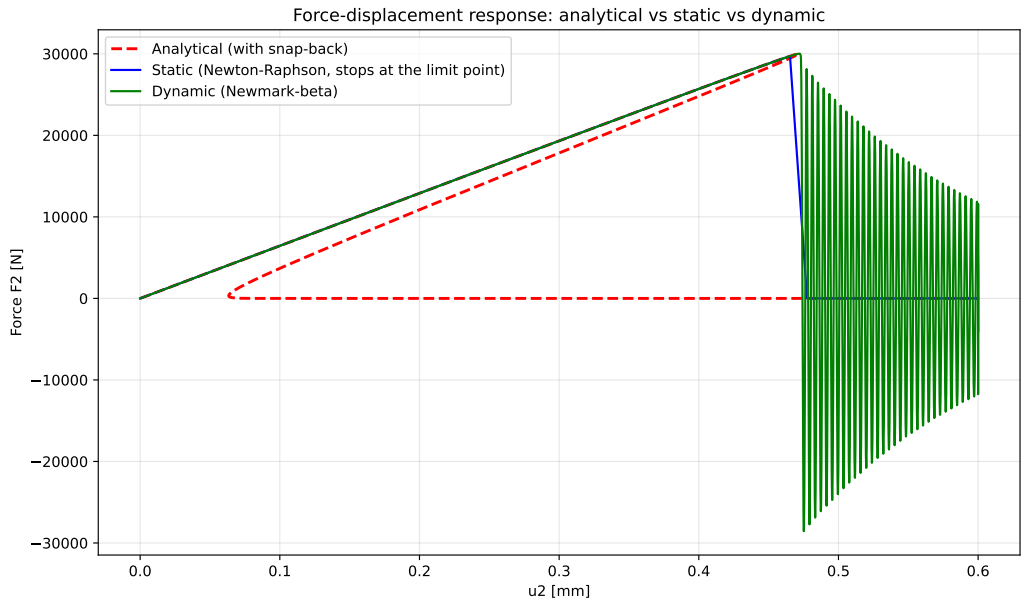


Figure 2: Force–displacement response $F_2(u_2)$: analytical (complete branch with snap-back), static (jumps or stops at the limit point) and dynamic (passes the limit point, then jumps and oscillates).

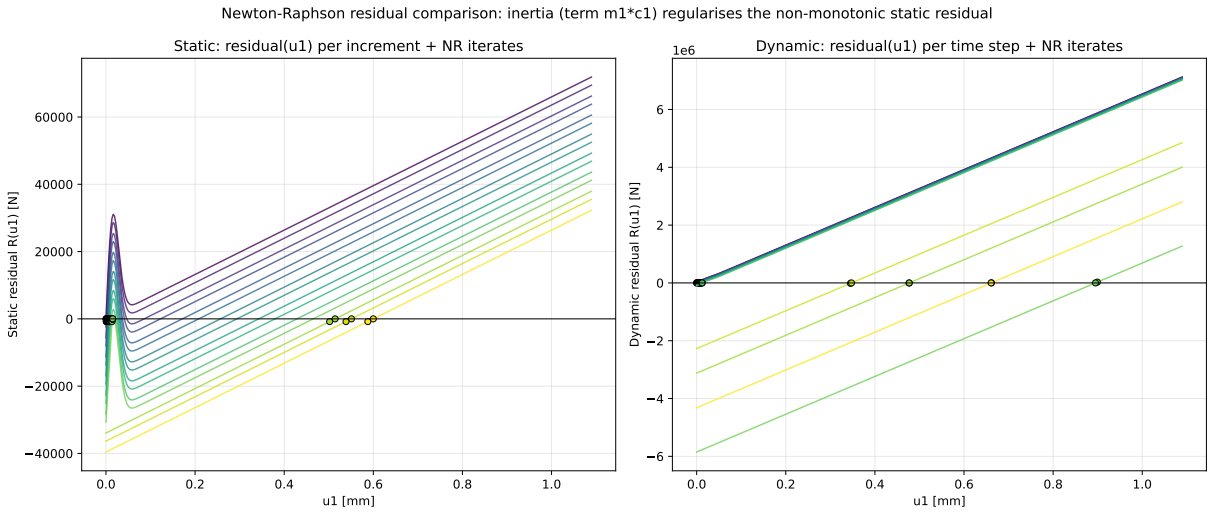


Figure 3: Residual $R(u_1)$ to be cancelled by the Newton–Raphson, one curve per increment (colour from purple, start of loading, to yellow, end) with the iterates as dots. Left: static, non-monotonic residual (bump due to the softening of spring 1), up to three roots. Right: dynamic, the term $m_1 c_1(u_1 - \tilde{u})$ makes the residual strictly increasing; the curves shift vertically with $\tilde{u}$, which carries the kinematic history.

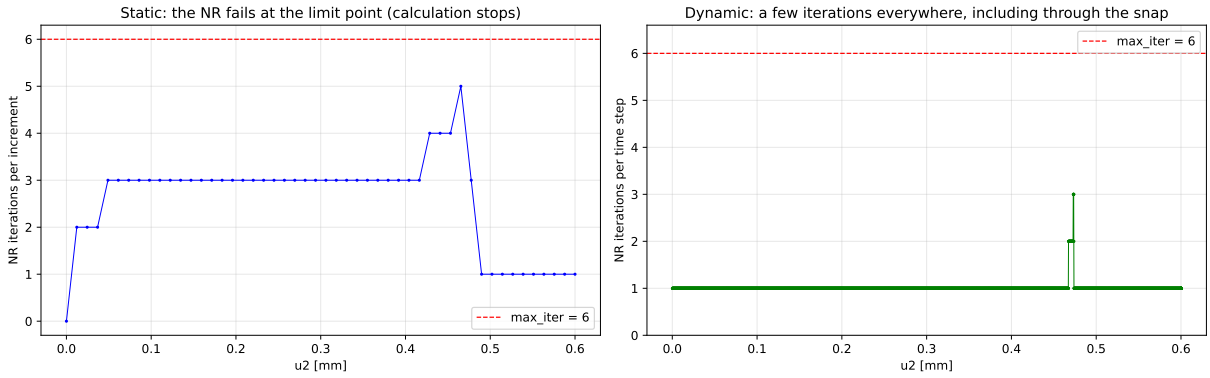


Figure 4: Number of Newton–Raphson iterations per increment (static, left) and per time step (dynamic, right). In statics the increment crossing the limit point is where the Newton–Raphson either jumps by accident onto the damaged branch or fails (`max_iter` reached and the calculation stops), depending on the increment size; in dynamics a few iterations suffice everywhere, including through the snap. This is the practical justification of the dynamic approach.

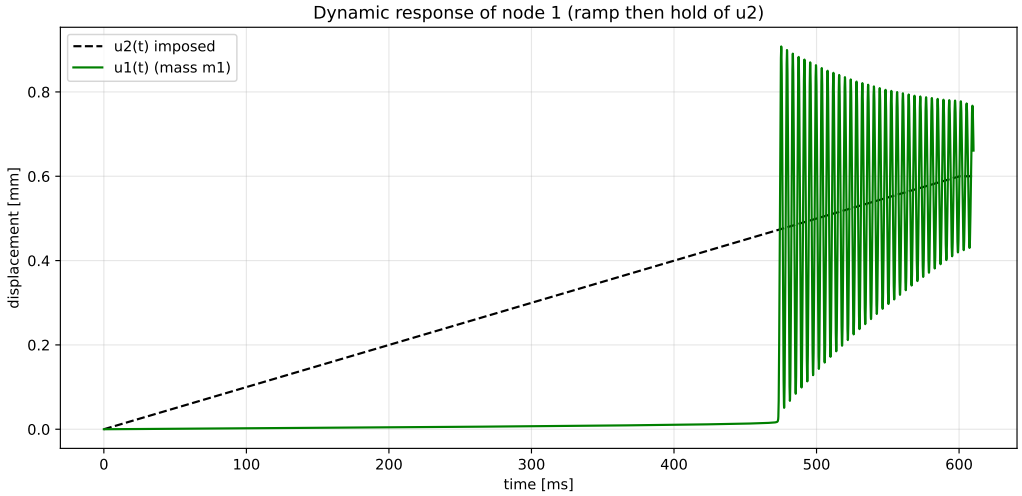


Figure 5: Dynamic time response: ramp of u_2 then hold; jump of node 1 and undamped free oscillation on spring 2.

7 Script layout and checks

`parameters.py`	physical and numerical parameters, derived quantities
`model.py`	damage law with irreversibility on d, static residual/tangent, Newton–Raphson
`analytical_solution.py`	branch (11), peak, energy balance
`static_solution.py`	u_2 increments, Newton–Raphson, iterate history
`dynamic_solution.py`	predictor–corrector Newmark, ramp + hold, iterate history
`comparison.py`	assembles the three solutions, figures (`-save` writes them to `figures/`)
`export_data.py`	writes every computed curve to `data/*.dat` (pgfplots, Manim)

Checks performed: numerical peak and limit point consistent with the analytical values; identity $\Psi_1 - \Psi_2 = \mathcal{D}_{\text{branch}} + |W|$ satisfied; Newton–Raphson convergence at each increment/step (tolerance 10^{-6} N; a tolerance of 10^{-9} N was below the round-off noise of the inertia terms $\sim 10^7$ N); positivity of the dynamic tangent; period of the free oscillations; conservation of $d = 1$ after the jump.

Limitations and extensions. Non-physical mass (regularisation parameter); no damping, hence perpetual oscillations after the jump (a damping $c\,\dot{u}_1$ or a dissipative Newmark $\gamma > \frac{1}{2}$ would make them decay); spring 2 is assumed massless; the static calculation does not follow the equilibrium branch beyond the limit point (an arc-length method would be needed for that).